

SARTHKI VACHHANI

Systems Programming | Full Stack Development | Python • C • JavaScript • FastAPI • React • Redis • PostgreSQL
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PROFESSIONAL SUMMARY

Computer Science undergraduate at PES University building production-grade systems across the full stack , from a Linux container runtime with kernel-level process isolation to a distributed task queue with Redis and PostgreSQL, and AI-powered full-stack platforms. Experienced in systems programming in C, backend development with Python and FastAPI, and frontend engineering with React. Passionate about scalable infrastructure, developer tooling, and applying computer science fundamentals to real engineering problems.

EDUCATION

PES University — Bangalore, India
B.Tech in Computer Science & Engineering | Expected 2028 | GPA: 7.7 / 10

TECHNICAL SKILLS

Languages:	Python, C, JavaScript, C++
Frontend:	React, Vite, HTML5, CSS3, JavaScript, NodeJS, NextJS, MERN
Backend:	Python, REST APIs, JSON/CSV Parsing, Data Modeling, ETL Pipelines
UI & Data Viz:	Streamlit, Matplotlib, Interactive Dashboards
Libraries:	Pandas, NumPy, Speech Recognition, OpenCV, Streamlit
Tools & Platforms:	Git, GitHub, Vercel, Ubuntu Linux, Kali Linux, FastAPI, Redis, PostgreSQL, Asyncpg
Core CS	Data Structures & Algorithms, Operating Systems, Computer Networks

PROJECTS

Multi-Container Runtime | C • Linux • Kernel Modules

- Engineered a Linux container runtime in C supporting PID, UTS, and mount namespace isolation, replicating core Docker primitives from scratch
- Implemented CFS scheduling experiments and memory limit enforcement via kernel modules, demonstrating low-level OS resource management
- Designed a supervisor process using pipes and UNIX sockets for IPC with concurrent structured logging across isolated containers

Distributed Task Queue | Python • FastAPI • Redis • PostgreSQL • React

- Built a distributed task queue from scratch in Python without any queue frameworks, implementing producer-consumer architecture with Redis as the broker and PostgreSQL for persistent job state
- Engineered atomic job dispatch using Redis RPOP to prevent race conditions across multiple concurrent workers, with automatic retry logic up to configurable max attempts
- Designed a REST API in FastAPI supporting job submission, status polling, and full job history, backed by an async PostgreSQL pipeline using asyncpg
- Built a real-time React dashboard displaying live job status, retry counts, and error tracking with 3-second polling

GigShield — AI-Powered Parametric Income Insurance (Ongoing) | React • Node.js • Python

- Building a full-stack insurance platform for gig workers using React, Node.js, and Python with trigger-based automated payout logic via real-time REST APIs
- Implemented ML-based fraud detection using Isolation Forest and graph-based anomaly detection to reduce fraudulent claim surface area
- Architected end-to-end data pipeline from event ingestion to payout decision, handling edge cases like disputed triggers and partial income loss

FinanceOS - SaaS Finance Dashboard | React • Tailwind CSS • Supabase • Recharts • Vercel

- Built a multi-user SaaS finance tracking dashboard with email authentication and real-time data persistence using Supabase.
- Implemented transaction management with category filtering, dynamic summary cards, and CSV export functionality.
- Visualized financial data through interactive monthly bar charts and spending donut charts using Recharts.

VoiceKind -Voice & Gesture Controlled Interaction System | Python • OpenCV • Mediapipe • SpeechRecognition • WebSocket • JavaScript

- Built a hands-free computer interaction system that maps index fingertip position to real-time cursor movement using Mediapipe hand landmark tracking and OpenCV, with moving average smoothing to reduce jitter.
- Implemented gesture-based click detection via closed-fist recognition and scroll control via vertical hand position zones, triggering PyAutoGUI actions.
- Integrated voice command recognition using SpeechRecognition library's Google backend with regex-based intent parsing for click, scroll, and navigation commands, bridged to a Chrome Extension via a local WebSocket server.

ACHIEVEMENTS

- 6th place — CyberClash Cybersecurity Hackathon (attack & defense using Kali Linux)
- Merit Scholarship Certificate